

Surface Tension & Critical Micelle Concentration

Capillary Rise Method across Three Liquid Systems · Distilled Water, Oil, and Aqueous SLS

GROUP 4

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What We Changed for Stage 3

Four deliberate upgrades to our Stage 2 protocol, each targeting a specific source of error or missing analysis.

01



Distilled water baseline

Replaced tap water with distilled water. Removes dissolved mineral contamination that was shifting our water surface tension above literature values in Stage 2.

02



Three liquid systems

Expanded from glycerol/surfactant mixtures to distilled water, oil, and SLS in water. Lets us compare a polar H-bonded liquid, a non-polar liquid, and an amphiphilic system side by side.

03



2 observations per reading

Every experimental observation was taken twice which enables statistical error analysis and error propagation on sigma, not just a point estimate.

04



CMC analysis added

Introduced Critical Micelle Concentration framework to interpret the surfactant result. Tells us whether our SLS concentration is above or below the point where micelles form.

The Physics: Surface Tension and CMC

A · SURFACE TENSION

What is Surface Tension

Molecules in the bulk are pulled equally in all directions. Molecules at the surface have fewer neighbours above them, so they feel a net inward pull. The surface contracts to the smallest possible area and behaves like a stretched membrane.

$$\sigma = F / L = dW / dA \quad [N/m = J/m^2]$$

- **Polar, H-bonded** High σ
- **Non-polar (Oil)** Lowest σ
- **Surfactant-laden (SLS)** Lower σ

B · CRITICAL MICELLE CONCENTRATION

Where surfactant behaviour flips

BELOW CMC

Surfactant molecules crowd the air–water interface with heads in water, tails sticking out. Each added molecule displaces water and lowers σ sharply.

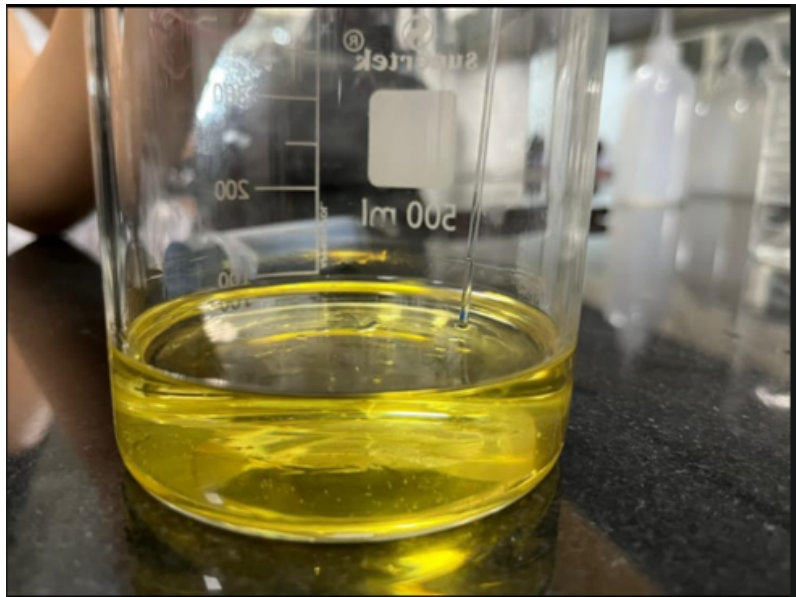
AT CMC

The interface is saturated. No more room at the surface. This is the critical concentration. σ stops falling.

ABOVE CMC

Extra surfactant self-assembles into spherical micelles in the bulk, with tails tucked inside and heads outside. σ stays flat no matter how much more you add.

Experimental Setup and Apparatus



APPARATUS

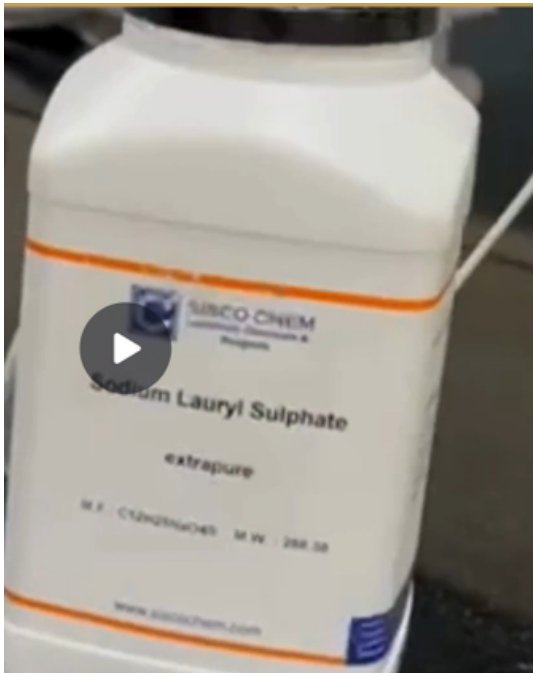
- Capillary tube I.D= 1.1-1.2 mm , O.D.=1.5-1.6 mm
- Beakers of varying volume
- Measuring scale · LC = 1 mm
- Distilled water, Cooking Oil, SLS (sodium lauryl sulphate)
- Iphone Camera , weighing scale

SYSTEM PARAMETERS

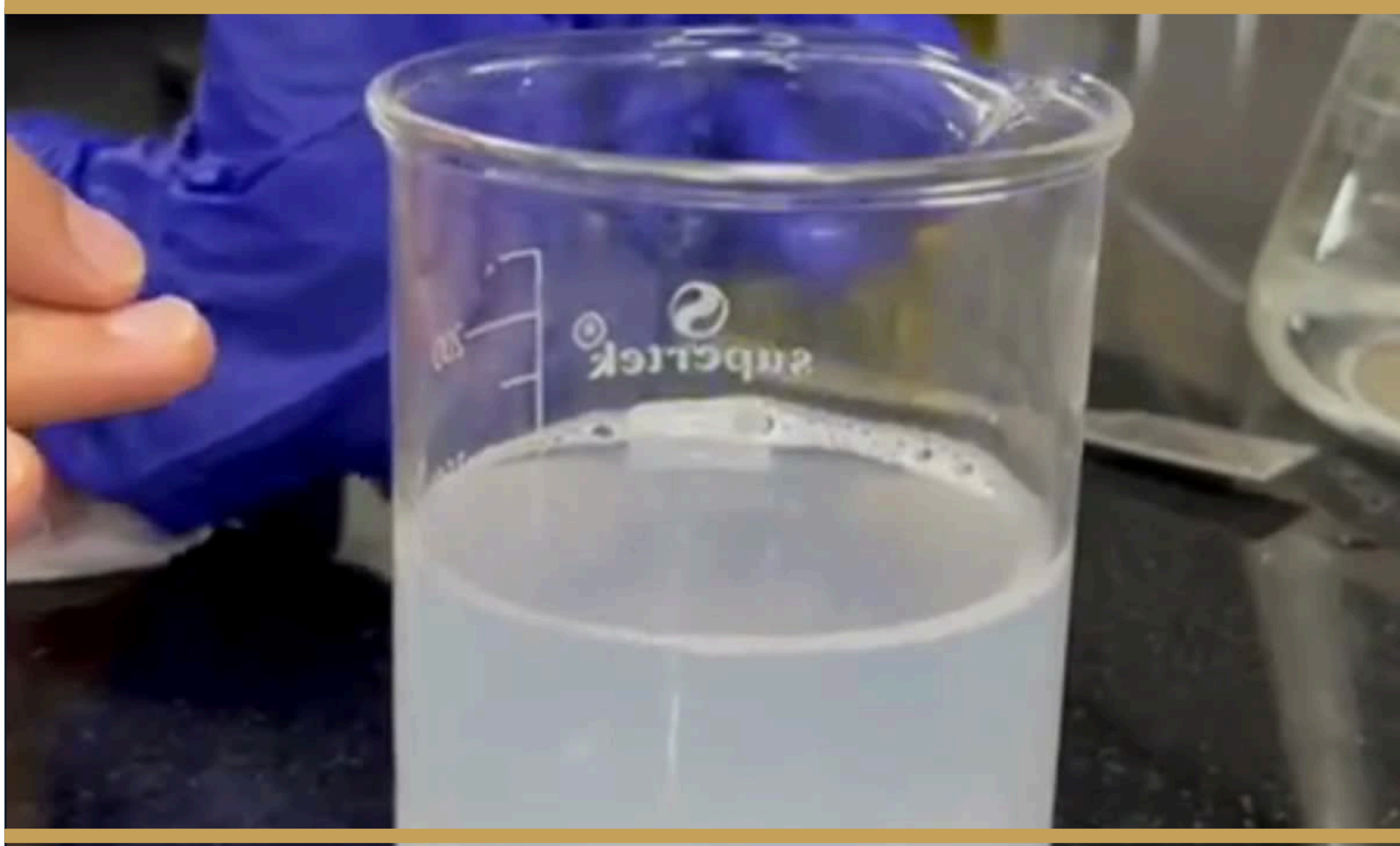
g	9.8 m/s ²
$\cos \theta$	≈ 1 (perfect wetting)
ρ_{water}	1000 kg/m ³
ρ_{oil}	900 kg/m ³
sls conc.	0.0133 g/ml

EXPECTED OUTCOME

$$\sigma_{\text{water}} > \sigma_{\text{SLS}} > \sigma_{\text{oil}}$$



Highlight: One-Minute Experiment Demonstration



Demonstration

1. This video shows our experiment using SLS(2g)+ distilled water(150 ml).
2. We put a capillary tube (clean and dry) in such a way that it just touches the surface of the solution.
3. Then we wait for 15-20 seconds.
4. Then we take the tube out gently and measure it using a measuring scale.

Capillary rise across SLS solution

Methodology and Governing Equations

A · FORCE BALANCE AT EQUILIBRIUM

Upward force (surface tension)

$$F\uparrow = \sigma \times 2\pi r \times \cos \theta$$

Downward force (weight of column)

$$F\downarrow = \rho \times g \times h \times \pi r^2$$

Setting $F\uparrow = F\downarrow$ with $\cos \theta \approx 1$

$$\sigma = \frac{1}{2} \cdot \rho \cdot g \cdot h \cdot r$$

B · PROCEDURE

- 1 Clean and dry the capillary tube thoroughly
- 2 Mount vertically, lower into the liquid
- 3 Wait ~ 20 seconds for meniscus equilibrium
- 4 Read rise height · repeat twice for every experiment
- 5 Compute σ , capillary length, Bond number

C · SUPPORTING DIAGNOSTICS

Jurin–da Vinci Constant

$$r \cdot h = 2\sigma / (\rho g)$$

Product of radius and height is constant for a given liquid. Confirms internal consistency across tube sizes.

Bond Number

$$Bo = \rho g r^2 / \sigma = (r / a)^2$$

Ratio of gravity to surface tension forces. $Bo \ll 1$ is required for the simple σ formula to remain accurate.

Raw Observations

Capillary radius : I.D.=1.1-1.2 mm , O.D.=1.5-1.6 mm · $g = 9.8 \text{ m/s}^2$ · Temperature $\approx 30^\circ\text{C}$ · Same observer throughout

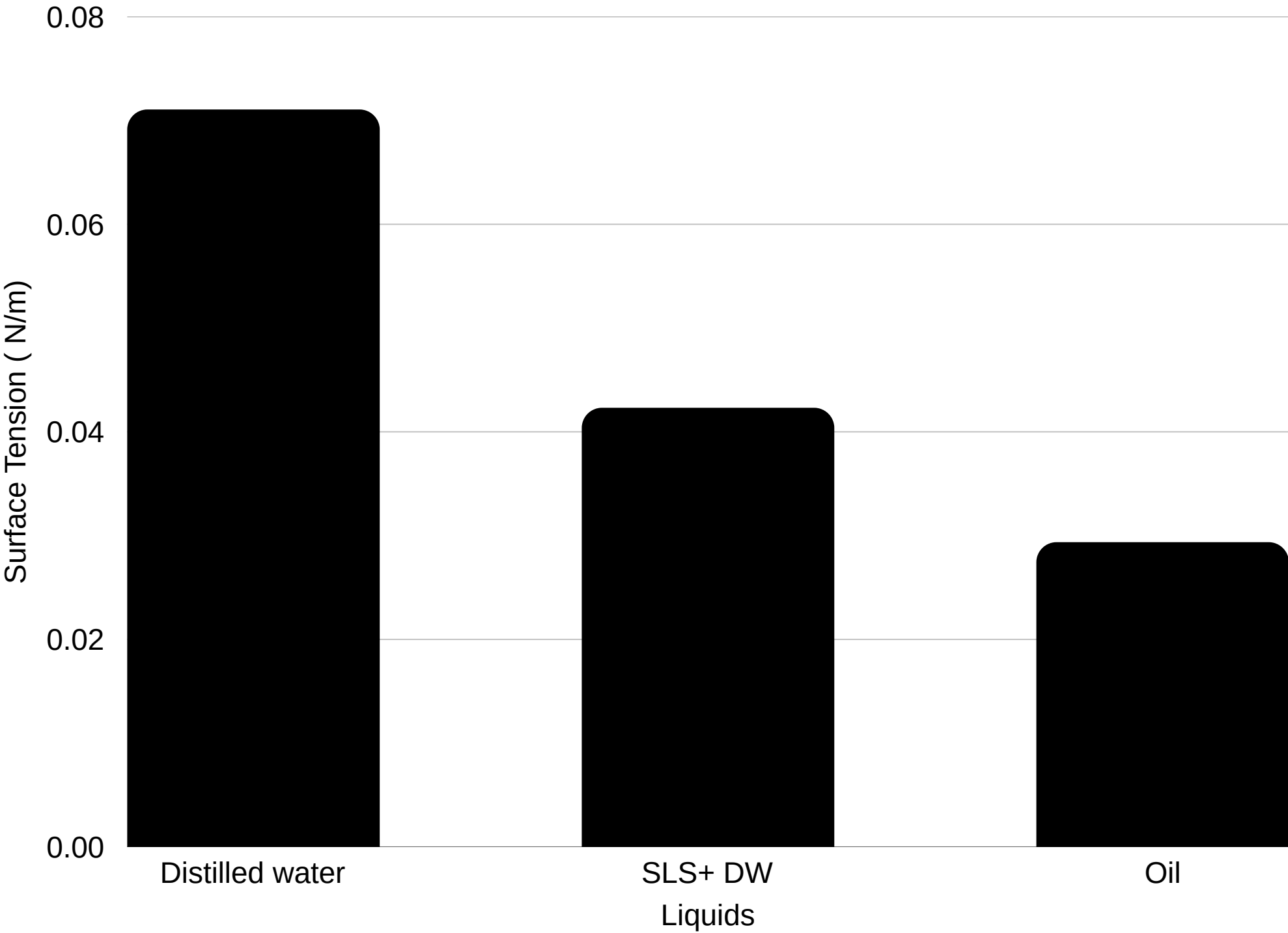
DISTILLED WATER				
$\rho = 1000 \text{ kg/m}^3$				
HEIGHT READINGS · cm				
50 ml	1	2.50	2	2.55
100 ml	3	2.55	4	2.60
150 ml	5	2.55	6	2.55
MEAN HEIGHT				
2.550 cm				
= 0.02550 m				
used in σ calculation				

OIL				
$\rho = 900 \text{ kg/m}^3$				
HEIGHT READINGS · cm				
100 ml	1	1.15	2	1.20
200 ml	3	1.15	4	1.20
300 ml	5	1.15	6	1.15
MEAN HEIGHT				
1.167 cm				
= 0.01167 m				
used in σ calculation				

SLS + DISTILLED WATER				
Concn.=0.0133 g/ml				
HEIGHT READINGS · cm				
2g SLS + 150 ml	1	1.50	2	1.55
2.66g SLS + 200 ml	3	1.50	4	1.55
3.33g SLS + 250 ml	5	1.50	6	1.50
MEAN HEIGHT				
1.517 cm				
= 0.01517 m				
used in σ calculation				

Results: Surface Tension and Diagnostics

SURFACE TENSION σ · N / m



DIAGNOSTICS PER LIQUID

Liquid	σ (N/m)	Height(cm)	Bond No.
● Distilled Water	0.07106	2.55	0.0225
● SLS + Water	0.04232	1.167	0.0378
● Oil	0.02937	1.517	0.0491

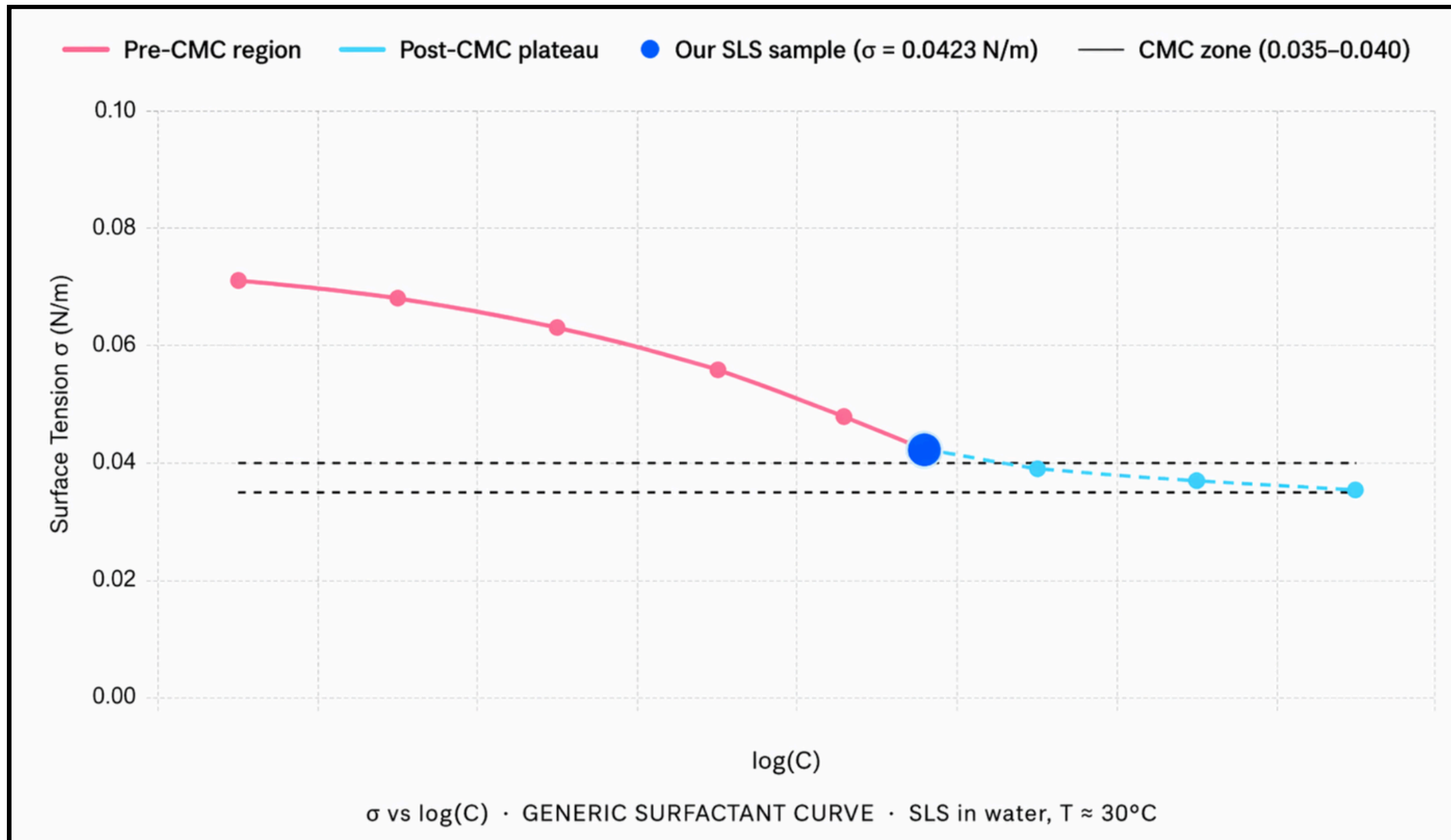
JURIN-DA VINCI CONSTANT · r × h			
WATER	SLS	OIL	
1.454 × 10 ⁻⁵ m ²	0.865 × 10 ⁻⁵ m ²	0.665 × 10 ⁻⁵ m ²	

KEY FINDING

σ order confirmed: water > SLS > oil

All Bond numbers below 0.05 means the simple σ formula is valid and meniscus distortion from gravity is negligible.

CMC Analysis: Where Does Our SLS Sit?



VERDICT

Our SLS is at or just above CMC.

$\sigma = 0.04232$ N/m sits just above the CMC plateau of 0.035–0.040 N/m. The interface is nearly saturated.

σ WATER

0.07106

N/m

σ SLS SAMPLE

0.04232

N/m

REDUCTION

40.5%

%

σ AT CMC

≈ 0.040

N/m literature

WHY THIS MATTERS

At actual CMC, σ drops to roughly 0.035 to 0.040 N/m. Our measured σ of 0.04232 is only 1.06× higher than that plateau, so the interface is nearly saturated. The solution is just short of forming micelles in the bulk — a small increase in SLS concentration would reach the CMC.

Note: This is a theoretical plot where we have mapped our experiment position to see where it lies around CMC

After adding 4g SLS in 250ml solution we didn't see any *significant height change* verifying our CMC region reached.

Error Analysis: Propagation and Precision

INSTRUMENTAL

± 0.05 cm

scale least count / 2

PROPAGATED

$\Delta\sigma = \sigma \cdot (\Delta h/h)$

from 2-reading sample std. dev.

PROPAGATED

$< 2 \%$

relative error on σ

PROPAGATED ERROR ON σ · PER LIQUID

LIQUID	σ (N/m)	σ from theory	PROPAGATED $\Delta\sigma$ (N/m)
● Distilled Water	0.07106	0.072	± 0.00051
● Oil	0.02937	0.03	± 0.00053
● SLS + Water	0.04232	0.045 (Approaching CMC)	± 0.00072

INTERPRETATION

All three relative errors sit under 2 percent. Combined with Bond numbers below 0.05, this confirms the simple σ formula is operating well within its valid regime. The measurements are precise enough to trust the σ ordering and the conclusion that SLS is at or near its CMC.

Conclusions and Report Snapshot

CHE-314 • COLLOIDS & INTERFACES • STAGE 3 FINAL REPORT

Measurement of Surface Tension of Liquid Solutions

Using the Capillary Rise Method across Three Liquid Systems

Distilled Water • Vegetable Oil • Aqueous Sodium Lauryl Sulphate

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COURSE ChE-314 Colloids & Interfaces
STAGE 3 (Final Evaluation)
DATE 21 April 2026
INSTITUTION Indian Institute of Technology Indore

KEY TAKEAWAYS

- ✓ **σ ordering validated**
Water (0.07106) > SLS (0.04232) > Oil (0.02937) N/m. Matches physical intuition around H-bonding, amphiphilic adsorption, and non-polar cohesion.
- ✓ **Distilled water fix worked**
Our water σ of 0.07106 N/m sits right on literature (~ 0.072 at 30°C). Confirms the switch from tap to distilled water removed the mineral contamination that was inflating Stage 2 values.
- ✓ **All Bond numbers well below 0.05**
Water 0.0225, SLS 0.0378, Oil 0.0491. The 1.05 mm capillary is firmly in the safe regime where gravity does not flatten the meniscus, so the simple σ formula stays accurate.
- ✓ **Relative error under 2 percent**
Six repeats per liquid tightened statistical Δh to the 0.02 cm range. Propagated $\Delta\sigma$ came out between ± 0.00051 and ± 0.00072 N/m across all three systems.
- ✓ **Our SLS sits right at CMC**
 σ dropped 40.5% from the water baseline, landing at 0.04232 N/m. That is only $1.06\times$ above the 0.035 to 0.040 N/m plateau, so the air-water interface is nearly saturated and the solution is at the onset of micelle formation.

References

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Thank You

All five members jointly performed every capillary reading. The following splits reflect who owned each post-experiment deliverable. Overall contribution is effectively equal.



Adya Singh Bishain

*Experiments, initial calculations,
initial reports and ppt, lit. Review and
final stage 3 calculations*



Soham Raut

*Calculations, experiments, content
review*



Mannan Mahajan

*Experiments, Calculations, making of
final report and presentations for the
2 stages*



Samarth Sharma

*Calculations, error propagation, Data
visualization*



Abhinav Singh

Filming, editing, voice-over